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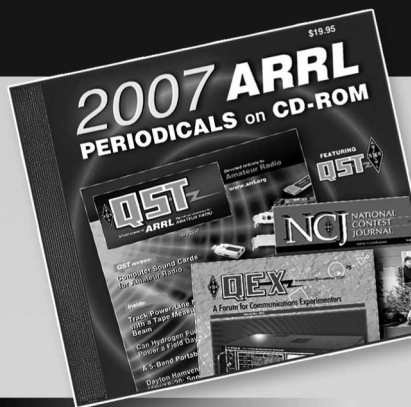
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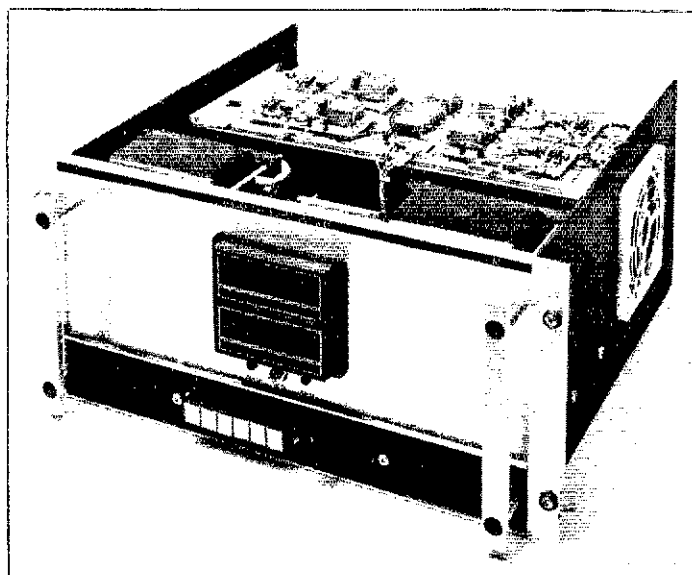
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MOSFET RF Power: An Update

Part 1: Power FETs are now practical for 2-30 MHz broadband-amplifier use at the kilowatt level. How do they compare with vacuum tubes and bipolar transistors? This report by ARRL TA Granberg provides some interesting answers.

By Helge Granberg,* K7ES/OH2ZE



Interested in power-FET technology? This paper combines a status report with some useful application notes for MOS power devices. It is intended to document the progress that has been made recently in power-FET development and deployment, but it is not meant as a construction article. The data presented are, however, suitable for amateur designers who wish to develop their own power-FET amplifiers.

Early power FETs were relatively low-level devices, but now these transistors can produce power amounts of 100 to 150 W. In contrast, we now have bipolar devices that yield up to 250 W of rf power (500-600 W in special water-cooled packages from at least one manufacturer). If we compare watts against dollars, using an 8877 tube as an example, a watt from a solid-state amplifier is about twice as expensive. If we include the necessary harmonic filters for broadband amplifiers, the price will be even higher.

Narrow-band solid-state amplifiers are considered feasible only for single-band use. This is because the low impedance levels (typical) make band switching the passive elements impractical. Also, losses will result.

Push-pull solid-state amplifiers aren't difficult to design, and they are desirable because the impedances are higher. Also, rf ground loops are easily eliminated. Other advantages are that the powers of the two devices are combined and the even

harmonics are suppressed. Broadband-transformer matching is suitable up to vhf. Other techniques, such as coaxial or other transmission lines (to provide a 180° phase shift) are practical up to the microwave spectrum.

The output harmonic filters of amplifiers usually are designed for 50 ohms, which makes them easy to switch. One low-pass filter usually covers less than an octave, but the frequency can be varied within the filter response without tuning or switching in a new filter. Because the amateur bands are one octave apart (except for 10, 18, 21 and 24 MHz), a separate filter is required for each band. More on this later.

Amplifier Specifics

The components for the circuit of Fig. 1 are not available from a single source. Many are engineering samples that were obtained from various manufacturers. As stated earlier, this treatment is conceptual rather than practical.

This amplifier provides a power output of 1600- to 1800-W PEP or cw, inclusive of the 0.3- to 0.4-dB filter losses, depending on the operating frequency. A nominal 40 W of drive is needed for full output. The input line contains an attenuator (selectable for 1, 2, 3 or 6 dB) to make the amplifier compatible with various commercial exciters, and to comply with FCC regulations. Over 2 kW of output power is possible with the 16 Motorola MRF150 MOS field-effect transistors used, but the power supply is rated only for 2800 W — the limiting factor.

The main power supply, shown in Fig. 2

(60 V no load, 48 V full load), consists of two smaller supplies. Each operates one of the large power modules. A regulated supply would dissipate some 400 to 500 W in the regulating process and would greatly increase the total weight of the system. A switching-mode supply would be rather complex for this power level, and would require RFI shielding.

Why FETs?

Power FETs have these definite advantages over bipolar transistors in this application:

- 1) More tolerant to load mismatch.
- 2) Simplified circuit design and biasing.
- 3) Lower high-order IMD (comparable to vacuum tubes).
- 4) Easier to make broadband because of higher input Z.
- 5) Gain can be controlled by varying the bias voltage. This can be used for alc shut-down instead of PIN-diode switches in the rf input. Linear alc can be had for ssb, but excessive bias reduction will deteriorate the IMD.
- 6) Higher power gain. The increase at 30 MHz can be 3 to 6 dB.

Industrial interest in power FETs probably relates to item 3. High-order IMD (9th order and up) causes adjacent-channel "splattering." This would also happen with an over-driven or mistuned tube amplifier. Low-order IMD (3rd and 5th) can be as high as -20 dB, and the signal will sound good if the high-order products are absent. Thus, the FCC specification is for only -25 dB on the 3rd-order product, and -60 dB or more for the 9th-order product and above (Marine).

*Motorola Semiconductor Products, Inc., Phoenix, AZ 85062

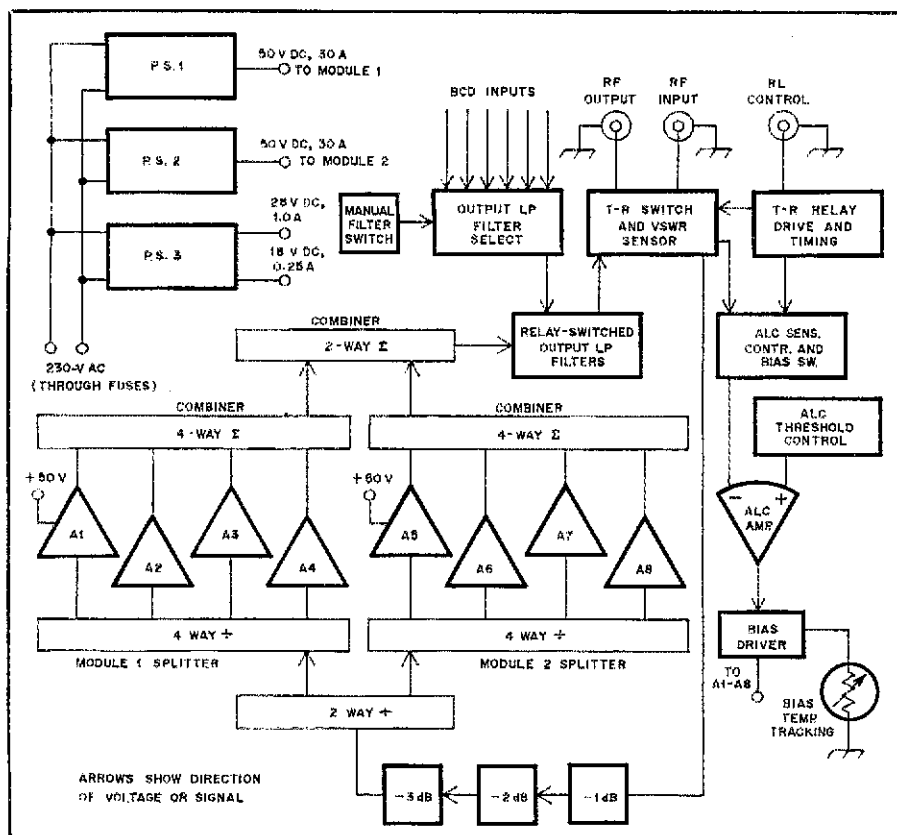


Fig. 1 — Block diagram of the 2-30 MHz power FET linear amplifier. A1-A8, inclusive, are individual FET amplifiers. See text for circuit description.

Bipolar transistors (depending on the type and internal structure) usually produce much more high-order IMD than is the case with FETs or tubes, unless the bipolar devices are biased to Class A and operated at reduced power.¹

The power gain of a common-source FET amplifier can be varied 20 dB or more by adjusting the bias voltage. All present rf power FETs are enhancement-mode MOS types: If the gate and source are at the same potential, there is no current flow through the drain. The gate of an N-channel device must be positive with respect to the source in order to "turn on" the transistor. The gain-control range depends on the initial gate-threshold voltage (1 to 4 V typical) and the amplitude of the voltage swing at the gate. The transistor will be turned off completely and become an attenuator if reverse bias is applied.

In this amplifier, the bias is lowered only to near ground potential — a gain reduction of 8 to 10 dB. This can be sufficient in the event of a shorted or open load condition. Protective control voltage is obtained from a reflectometer at the amplifier output (Fig. 1). This amplifier also includes a self-contained linear a/c system. A/c voltage is not fed back to the exciter as is done normally. The control range is limited to about 2 dB, beyond which it would degrade the IMD ex-

cessively. Bipolar-transistor amplifiers usually employ a variable voltage or current attenuator at the input port. PIN diodes are often used, but for linear a/c an elaborate circuit is needed to prevent harmonics and distortion.

General Description

A block diagram of the total system (exclusive of the digital panel meters) is shown in Fig. 1. One meter reads 0-199.9 V and the other is for 0-199.9 mV. The latter one reads current and has 50-A and 50-mA shunts. The shunts (3) are shown in Fig. 2 as 0.001-ohm resistors. The shunts are used to read the individual currents of the rf modules, and one shunt is utilized to monitor the total current. The voltmeter monitors the nominal voltages of the main supplies, or about one half the voltage difference when switched to both supplies.^{2,3} This metering system eliminates the need for heavy-duty wiring and prevents large currents from flowing through the switches.

There are two main rf modules (Fig. 1). Each contains four push-pull amplifiers. Combiners are used to produce a summed output of 1600 to 1800 W. Two pieces of Aavid Engineering heat-sink extrusion (no. 60140) support the amplifier boards, the combiners and other components of each module. Four copper heat spreaders (each 3/16 × 2 × 5-1/2 inches; mm = 25.4 × in.) are mounted individually on the flat surfaces of the heat sinks. The

layout permits four MOSFETs (two boards) to be mounted on each of the heat spreaders. Copper is nearly twice as good as aluminum for heat conduction. Hence, it improves the instantaneous heat transfer by spreading the concentrated heat from the transistor flanges more efficiently along the aluminum heat sinks. The two heat sinks are supported by means of aluminum plates on each side. The plates also serve as mounting brackets for the whole structure. This technique provides a channel with the heat-sink fins inside. Two 5-inch fans (actuated by two 75° C thermostats — normally open) force air through the channel. The thermostats are attached to the heat sinks. During cw operation, the fans cycle about two minutes on and five minutes off.

Fig. 3 shows the component layout. Near the top is one of the power modules with four amplifier boards. The output combiner is shown at the center of the module. The lower part of the assembly contains the power supply. At the upper left are the two- and four-port splitters. The two-port main combiner is at the upper right.⁴

The transistor leads are pressed down (not soldered) against the pc boards and related contact areas. Teflon rings, then silicone-rubber rings, followed by aluminum rings, ensure firm contact when pressed in place by means of special standoffs. This method makes field service easier when replacing a transistor, since no soldering is required.

Filtering and T-R Circuit

The main combiner output is fed to a bank of low-pass filters. These are relay-switched for the desired band. The front-panel control switch operates when the BCD inputs to the filter-select circuit are open or high. When one or more of the BCD inputs are grounded, the manual switch is disabled and a light indicates the filter that has been selected by the code. This feature is useful for automatic band changing with transceivers that are designed for computer control.

Output from the filters goes to a T-R switch that consists of two Kilovac HC-1/530 vacuum relays. One is located at the amplifier input and the other is at the amplifier output. The relays are housed in separate shield enclosures (with BNC interconnect) to minimize unwanted crosstalk. T-R relay timing and drive signals must occur in a precise timing sequence to prevent "hot switching" in the amplifier output. Thus, the output relay must be energized first and released later than the input relay. For full QSK, the delays should be minimized. In this circuit the limit is about 8 ms, owing to the speed of the relays. Longer delays would hardly affect the QSK operation, but would shorten the marks for RTTY and cw, which would be apparent at high operating speeds. The control signal from the transceiver requires a key-to-ground

¹Notes appear on page 16.

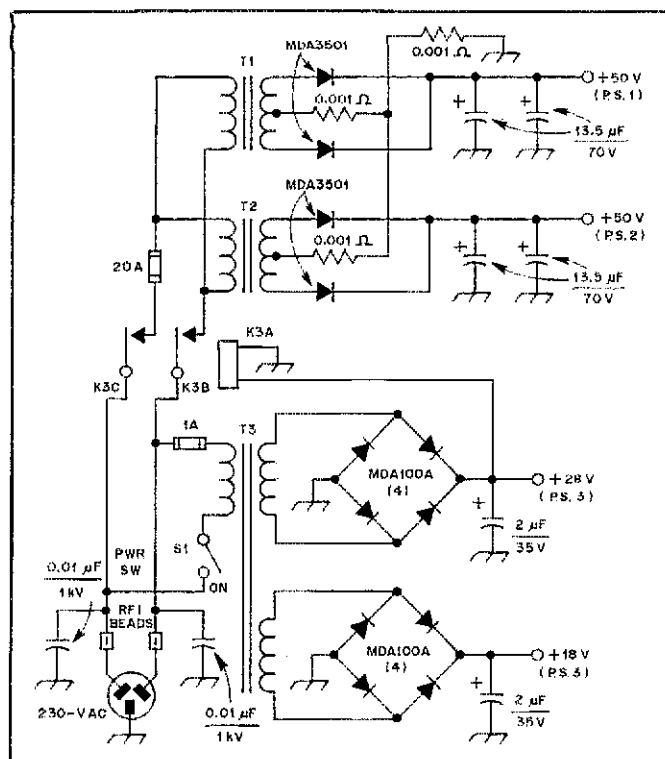


Fig. 2 — Schematic diagram of the FET amplifier power supply. Capacitors are disc ceramic, except those with polarity marked, which are electrolytic. Z1 and Z2 are ferrite beads for RFI suppression (Fair Rite no. 2673021801).

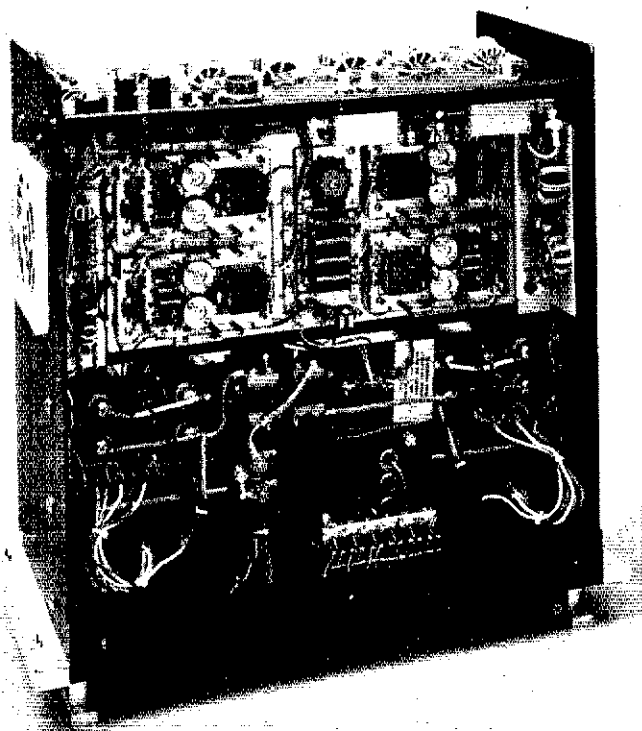


Fig. 3 — The inside of the solid-state amplifier. See text for discussion of module positioning.

polarity. The T-R, timing and drive-circuit module is visible at the upper right in Fig. 3 (just under the rear panel).

A reflectometer type of VSWR sensor is housed in the T-R circuit enclosure. Only the reflected power is measured. Output is routed to the circuit board that contains the alc, bias-temperature tracking and automatic filter-select circuits. Regulators for 12 and 24 V are also located on this board. Ferrite beads are used on the leads that enter this board. This prevents rf from getting into the alc amplifier and CMOS logic circuits.

The T-R output relay control voltage is routed also to the alc circuit. This turns off the transistor bias during standby, thereby preventing the 400-W standby dissipation (500 mA per transistor) of the 16 devices.

The bias-temperature tracking feature keeps the idling current constant with increasing heat-sink temperature. Normally, it would approximately double from 25° C to 75° C. This function is handled by a thermistor that is coupled to one of the heat sinks. Idle-current variation is 20% or less.

Circuit Details

Fig. 4 contains a schematic diagram of one of the push-pull amplifiers. The circuit is much simpler than one with bipolar transistors. The external resistor values

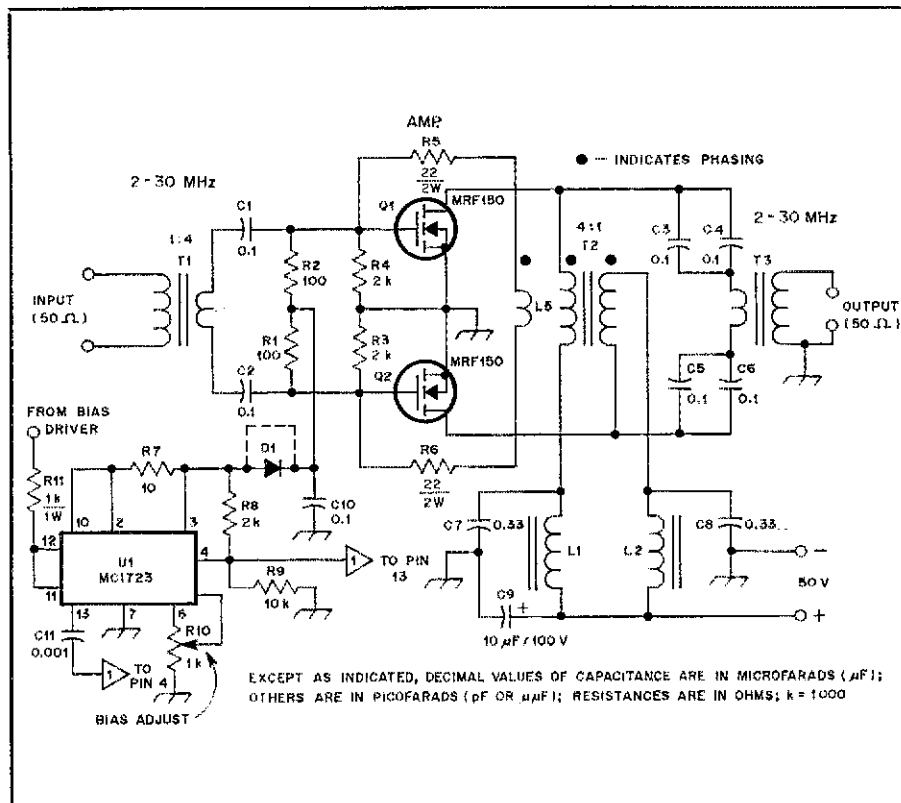


Fig. 4 — Schematic diagram of one of the power modules, inclusive of the bias regulator. C1, C2 and C10 are monolithic capacitors. C3-C8, inclusive, are ceramic chip capacitors. C9 is electrolytic and C11 is disc ceramic. Fixed-value resistors are carbon-composition, 1/2 W, unless noted otherwise. R10 is a Trimpot®. See text for discussion of the remaining components.

for the bias regulator permit regulation of the load but not the input voltage. The regulator serves three purposes: (1) Provides convenient bias adjustment (0.5 to 9 V) with R10; (2) provides a current sink for fast discharge of C10, for alc shut-down; (3) gives isolation between the bias circuits of each amplifier to the common-base driver. Hence, the adjustment of one amplifier does not affect the bias levels of the remaining amplifiers. D1 was used on the initial test board for protection of the regulator. It should be jumpered out for this application to prevent defeating the fast alc action. The regulator of Fig. 4 switches off in about 120 μ s when the voltage from R11 drops from 24 to 2 or less. Considering all of the time constants and delays in the loop, the alc can still react in less than 0.5 ms.

The gates of Q1 and Q2 present an almost pure capacitance, respective to the sources, at 2-30 MHz. To ensure stability, the high input Q is lowered by means of R1, R2, C10 and negative feedback from T2. The source impedance and feedback are controlled by R5, R6 and L5. T2 is wound on a TV-antenna-balun style of core with a bifilar winding. L5 consists of one turn on the same core. The ferrite core should have a μ_c of at least 800 and a high Curie temperature for use down to 1.8 MHz. Teflon-insulated wire is recommended because T2 can reach temperatures in excess of 150° C. T1 and T2 are the common ferrite sleeve/metal tube style of transformers described in note 2. The transformation ratios are 4:1 and 1:4, respectively. A low-loss ferrite core (Stackpole C7/DB) is used for T3. It was chosen to prevent core overheating during extended periods of operation. It is available in a rectangular balun format (no. 55-7051) and has a larger cross-sectional area than the more common no. 57-3238 ferrite sleeves.

Most failures during initial testing occurred from overheating in the output blocking capacitors. Since they are ceramic chip capacitors, soldered rigidly to the pc board, cracks appeared in them. These capacitors must handle an average rf current of 4 to 5 A, but at relatively low voltage. Paralleled disc-ceramic capacitors were tried at C3-C6, inclusive. They worked fine, but were bulky for this board layout. They serve to dc-isolate T3 (unnecessary with this style of transformer) and to compensate for the frequency versus output-impedance slope of the transistors. Part 2 of this paper will appear in a subsequent issue of QST. 

Notes

"Power MOSFETs versus Bipolar Transistors," AN-860, Motorola Semiconductor Products, Inc.

The Radio Amateur's Handbook, 59th edition (Newington: ARRL, 1982).

DC Power Supply Handbook (Application Note 90A), Hewlett-Packard Co.

"Broadband Transformers and Power Combining Techniques for RF," AN-749, Motorola Semiconductor Products, Inc.

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